

# JIANKE YANG

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## EDUCATION

**UNIVERSITY OF CALIFORNIA SAN DIEGO**, San Diego, CA Sep 2022 - Present

PhD in Computer Science

- Advisor: Prof. Rose Yu
- Research area: Geometric Deep Learning, Generative AI, Physics-Guided Deep Learning

**TSINGHUA UNIVERSITY**, Beijing, China Sep 2018 - Jun 2022

Bachelor in Computer Science and Technology; Minor in Statistics

- Major GPA: 3.94/4; Rank: 5/231

## PUBLICATIONS

**Jianke Yang**, Manu Bhat, Bryan Hu, Yadi Cao, Nima Dehmamy, Robin Walters, and Rose Yu. Discovering Symbolic Differential Equations with Symmetry Invariants. *Preprint*, 2025.

**Jianke Yang**, Wang Rao, Nima Dehmamy, Robin Walters, and Rose Yu. Symmetry-Informed Governing Equation Discovery. *Neural Information Processing Systems (NeurIPS)*, 2024.

**Jianke Yang**, Nima Dehmamy, Robin Walters, and Rose Yu. Latent Space Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2024.

**Jianke Yang**, Robin Walters, Nima Dehmamy, and Rose Yu. Generative Adversarial Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2023.

Manu Bhat, Jonghyun Park, **Jianke Yang**, Nima Dehmamy, Robin Walters, and Rose Yu. AtlasD: Automatic Local Symmetry Discovery. To appear at *International Conference on Machine Learning (ICML)*, 2025.

**Jianke Yang**, Honglu Zhou, Kai Li, Rose Yu, Martin Renqiang Min. Physics-Informed Human Motion Generation. *In submission*, 2024.

## WORK EXPERIENCE

**Research Intern, Bytedance Inc.** Jun 2024 - Sep 2024

- Latent diffusion model for protein dynamics generation

**Research Intern, NEC Laboratories** Jun 2023 - Sep 2023

- Diffusion model for human motion generation
- Incorporate physics simulator into training to perform physically valid generation

## TEACHING ASSISTANTSHIP

**CSE291 Generative AI**, UC San Diego Sep 2023 - Dec 2023

## INVITED TALKS

- (Mar 2025) **Boston Symmetry Day**, Northeastern University
- (Nov 2024) **AAAI Fall Symposium** on Computational Approaches to Scientific Discovery
- (Oct 2023) **SciML Webinar**, University of Michigan

- (Mar 2023) **ML in Physics (Guest Lecture)**, UC San Diego

## **HONORS & AWARDS**

- Academic Excellence Award (2019, 2020, 2021) at Tsinghua University
- First Prize (2020) in China National Contest of Mathematical Modeling for Undergraduates

## **PROGRAMMING SKILLS**

Python, R, MATLAB, C/C++, Java, JavaScript, Rust